U.S. Army

Integrated Personnel and Pay System–Army (IPPS-A) Program Management Office (PMO) Configuration Management Plan (CMP)

Version 8.0

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Signature Page

Integrated Personnel and Pay System–Army

Configuration Management Plan

Version 8.0

CONCURRENCE / APRROVAL

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8.0	14 February 2025	Updated Post Go-Live Integrating Agile Practices

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1.0 INTRODUCTION

1.1 Purpose

The Integrated Personnel and Pay System–Army (IPPS-A) Configuration Management Plan (CMP) describes the Configuration Management (CM) functions and provides guidelines and structure to IPPS-A Program Management Office (PMO), Functional Management Division (FMD), Stakeholders, and System Integrator (SI) personnel involved in CM activities. CM tasks include Configuration Item (CI) identification, control, tracking, and accounting for Sustainment maintenance. CM also includes changes to IPPS-A Production System (infrastructure, software, and documentation).

1.2 Scope

The IPPS-A CMP is applicable to all IPPS-A PMO staff (civilian, military, and support contractors), the SI, and applicable stakeholders. All CM processes and procedures apply to both the Unclassified and Classified products and environments.

1.3 Changes to Document

The Government Release / CM Manager is responsible for the maintenance of this plan. This CMP is approved as of the date signed and remains in effect until superseded by a newer version.

This plan will be reviewed annually at a minimum and updated as required throughout the program's lifecycle to reflect changes in CM activities. All changes will be in effect upon approval.

The SI CM responsibilities include:

- Method for implementing and maintaining version control of code builds
- Processes and procedures for the operation and use of Government-approved CM and CI
- Build and migration tool(s) for PeopleSoft and Third Party Applications

The following work products are due to PMO as outlined in Table 1-1.

Additional revisions are due ten (10) business days after receipt of Government comments provided during the review meetings noted in the table below.

Table 1-1 System Integrator CI Update Frequency

Configuration Item (CI)	Review Meeting	Frequency
13-009	Configuration Management Plan	Annually
13-010	Quarterly Configuration Status Accounting Information (CSAI) Report	Quarterly
13-011	Configuration Item Document Updates	Bi-weekly
13-016	Program Metric Reports	Monthly
13-017	Release Notes	Bi-weekly

1.4 Background

The objective of IPPS-A is to deliver a single integrated system, which enhances the efficiency and accuracy of Army personnel and pay procedures. IPPS-A provides personnel and pay management functionality for the Army (Active, Guard, and Reserve).

IPPS-A development is executed using the Agile SAFe methodology to produce builds of system functionality, adding new capabilities with each subsequent release.

The IPPS-A System provides:

- Single Comprehensive Personnel and Pay Record
- Self-Service Access
- Data Standardization
- Integrated Personnel and Pay

- Talent Management
- Strength Management and Accountability
- Personnel Asset Visibility
- Efficiencies Gained by Workflow
- Cross Component Support
- Business Process Standardization
- Auditability
- Total Force Readiness
- Non-secure Internet Protocol Router Network (NIPRNet) and Secret Internet Protocol Router Network (SIPRNet) Integration with Cross Domain Support
- Army Military Pay (Future Capability)

2.0 CONFIGURATION MANAGEMENT

The objective of IPPS-A CM is to ensure all Continuous Integration (CI), e.g., requirements, design and test artifacts, infrastructure, and software components, are under configuration control to ensure the integrity of the system. Collaborate with the Sustainment Support Team, serve as a resource, and assist in developing and implementing the PMO CM processes and procedures. These processes will aid in tracking of CM and action items, management of the change repository, and preparing and coordinating CM correspondence and documentation. The CM Team will support the following CM activities:

- Support the development and maintenance of the Government's CM policy, plan, charters, processes, and procedures
- Review CM plans for compliance with Software Engineering Institute/Capability Maturity Model
- Assist in monitoring configuration control and configuration status for conformance with the Military Handbook-Configuration Management Guidance
- Provide configuration accounting in support of change management activities and facilitate or participate in applicable Control Boards
- Participate in audits of products and processes implemented by other PMO contracts
- Participate in Lab/Test Readiness Reviews
- Use the Government's CM tool to maintain a repository of, track, and report on Engineering Change Request (ECR), Change Requests (CR), Incidents and baseline analyses
- Monitor and assess the effectiveness of the CM effort
- Configuration Management Library

The CM process is applied throughout the project's lifecycle to accomplish the following:

- Identify, define, record, and baseline the product and work product definitions
- Control their modification and release
- Record and report the status of modification requests
- Ensure the continuing integrity of the project definition and working environment

CM consists of five major areas: CM Planning & Management, CM Identification, Change Management (i.e., Configuration Control), Status Accounting, and Verification & Audit.

2.1 Configuration Management Methods & Tools

All CM-related IPPS-A program management tools are interrelated to support ease of use and ensure accuracy by entering data once and creating reports from an approved database. Below are the current program management tools used to support CM activities. APPENDIX B contains additional tool details.

Table 2-1 CM Methods and Tools

Process	Tools
Requirements Change Management	Azure DevOps (ADO)
Documentation Change Management	SharePoint
Release Management – PeopleSoft	Phire
Configuration Control – CI Tracking	Service Now (Discovery)

3.0 CONFIGURATION MANAGEMENT ACTIVITIES

This CMP addresses all specific and generic goals and practices identified for the CM process area.

3.1 Configuration Planning and Management

3.1.1 CM Organization and Resources

Both the PMO and the SI are responsible for coordinating, implementing, and maintaining CM disciplines within IPPS-A. The following sections provide insight into the PMO organization's contribution. The implementation of the SI's role in the CM process is discussed in SI's CMP (WP CDRL 13-009).

3.1.2 Product Manager

The IPPS-A PdM is committed to maintaining a disciplined CM program by providing management, direction, and guidance throughout the project life cycle. Specific PdM responsibilities include:

- Approve the parameters for the Technology Control Board (TCB), and Collab
- Ensure that PMO staff supports CM activities
- Oversee the development and maintenance of the IPPS-A program
- Approve any Policy Memoranda or any compliance-related documentation that impacts the CM activities

3.1.3 Chief, Applications Division (Apps Div)

The Chief, (Apps Div) is responsible for ensuring that the TCBs and Collabs are conducted according to the CMP. In the absence of the Apps Div Chief, this role is delegated to the Government Release Manager and is supported by a Contractor CM Lead. Specific responsibilities are:

- Define the parameters for the Technology Control Board (TCB), and Collab
- Chair the TCB sessions
- Provide primary vote for ECRs in the TCB and Collab
- Ensure that the TCB is conducted according to the CM Plan
- Review any Policy Memoranda or any compliance-related documentation that impacts the CM activities
- Advise the PdM on any CM-related decisions
- Oversees the establishment and maintenance of configuration version control of IPPS-A baselines

3.1.4 Release / CM Manager

The Release / CM Manager is responsible for assuring that the IPPS-A PMO is complying with the PEO EIS Policy Memorandum 10-17, Configuration Management. The Release / CM Manager is also responsible for coordinating and managing all IPPS-A CM activities. PEO EIS Policy Memorandum 10-17 is provided to the Release / CM Manager for review and concurrence of compliance.

The Release / CM Manager ensures that all ECRs are appropriately submitted and discussed between FMD, the SI, and PMO Technical and Functional leads. Specific responsibilities include:

- Participating in and supporting the TCB and Collab
- Producing, maintaining, and implementing the CM Plan and processes that define the workflow for all CM activities
- Overseeing the day-to-day CM activities, including oversight of the SI's CM function
- Obtaining adequate training for CM and project staff to perform CM activities
- Maintaining configuration control for all baselined program CIs
- Ensuring the implementation of CM tools meets CM requirements
- Confirming adherence to contractual and data requirements for CM in the form of CDRL reviews
- Maintains configuration and version control of IPPS-A baselines
- Participates in configuration audits and reviews

3.1.5 CM PMO Support Lead

Specific CM PMO Support Lead responsibilities include:

- Facilitates and supports the Defect Review Board (DRB), and TCB
- Prepares the ECRs, with participation and contribution by the FMD and the SI, as needed, that will be collaborated among the Functional and Technical Teams
- Participates in configuration audits and reviews
- Maintains the CM Library
- Reviews the CSA reports and maintaining CSA records, per the requirements of CDRL 13-010
- Maintains configuration and version control of IPPS-A baselines
- Familiarization with the CM program documentation and tools
- Attending CM training, as required
- Participates in the DRB, TCB, and Collab, as required
- Updates the ECRs per the ECR process

3.1.6 Program Control Boards

The Program Control Boards include the DRB, TCB and Collab. Table 3-1 describes the purpose of each Control Board, as well as the Members and Chair representatives.

Table 3-1 IPPS-A Program Control Boards Impacting Changes

Board	Purpose
DRB	• Reviews and dispositions all Incidents, Service Requests (SRs), Engineering Change Request- Lows (ECR-Ls) and System Problem Reports (SPRs) • Determines when an SPR is not a defect and should be processed as an ECR
TCB	The Technology Control Board (TCB) is the decision board for: • New requirements adjudication, acceptance, prioritization, for Engineering Change Request (ECRs) and modifications against the approved Functional Baseline • Any change escalated from the DRB for resolution • Infrastructure updates, maintenance and patching • Establishment of the Allocated and Product Baselines • Specification changes and other artifacts that may impact program cost, internal schedule, and performance • Hardware installation/changes

3.2 Configuration Identification

Configuration Identification is the process of identifying items to be placed under formal configuration control, in support of recording the technical description of a system and its discrete components through lifecycle.

Once an item is formally designated as a CI, it must be placed under configuration control in a system that provides version and access control.

Configuration Identification results in approved configuration documentation, establishment of baselines, creation of records in a status accounting database, and documentation to support configuration verification and audit.

The following always applies to Configuration Identification on the program:

- The CM Team reviews the CIs and their relationship to requirements and baselines
- Use of the CI structure throughout the IPPS-A lifecycle facilitates effective support of items in service

CM guidelines are in place, and that the Government PMO, FMD, the SI, any applicable SI subcontractor, and any other Government-contracted effort have configuration documentation change control responsibility for each artifact.

CIs selected at a level that benefits all stakeholders, but that does not create an administrative burden.

3.2.1 Baselines

The SI CM Team is responsible for configuration control of all IPPS-A system baselines and products. Baselines are established and controlled by the CM Team for each IPPS-A Release. The Functional Baseline establishes the high-level requirements to achieve a successful, useable, and effective system.

Creating a baseline involves the assignment of an identifier (i.e., label) that is used to control the configuration of an item and its associated entities.

Releasing a baseline constitutes retrieval of labeled files (the CIs) for deployment. The following items, when assigned a unique identifier either individually or collectively as a set, can be considered to be a baseline, that defines a formal release or version of the system:

- Functional requirements
- Design specifications
- Source code files and the associated executable code, build files, and user documentation

A separate baseline is created in the program for each major or minor release, as defined in the

Release Management Plan (RMP) and within the release for each acquisition milestone event.

Establishing and maintaining configuration baselines ensures that only approved changes are made to the requirements, design, and components of IPPS-A.

3.2.1.1 Establishing Baselines

A baseline is in its essence a snapshot of the state of an item or system at a point in time. The SI CM Team is responsible for maintaining and managing the IPPS-A baseline.

3.2.1.2 Establishing System Baselines

The method of identifying system level baselines for the IPPS-A is the responsibility of the SI CM Team.

3.2.1.3 Requirements Baseline

A Requirements Baseline will be used to identify the system functional, technical, and security requirements. The SI CM Team is responsible for maintaining this baseline. Changes to the requirements baseline are identified and confirmed at the TCB.

3.2.1.4 Functional Baseline

The IPPS-A Functional Baseline is managed and monitored by the SI CM Team. Functional Baseline CIs are stored in ADO and placed under formal CM.

3.2.1.5 Allocated Baseline

The Allocated Baseline is established, monitored, and managed by the SI CM Team for each release and was finalized at the Critical Design Review (CDR). Any change to the Allocated Baseline must be approved by the TCB through the ECR process. The Allocated Baseline CIs are stored in the applicable repository (ADO or SharePoint), dependent upon the type of CI and are under formal CM.

3.2.1.6 Production Baseline

The initial IPPS-A product baseline was established by the SI CM Team at the CDR.

Table 3-2: Product Baseline Configuration Items

Product Configuration Item
Configuration Status Accounting Report
Requirements Traceability Verification Matrix
Information Assurance Vulnerability Matrix
Release Plan
System Version Description
Design Documentation
Data Dictionary
Database Schema
Data Model
Technical Architecture Documentation

Product Configuration Item
Training Documentation
Operations & Maintenance Manuals

3.2.1.7 CIs Contained in a Baseline

All changes to the production system will occur under formal configuration control procedures that include Control Board authorization, information security authorization, and a disciplined release management process.

3.2.1.8 Publishing Baselines

Baseline reports will be provided as standard output products of ADO and SharePoint, used among other CM tools on the program.

3.2.2 Naming Standards

CIs should be named in a logical way that traces back to the highest-level CI but is descriptive enough to capture information about the object or element after which it is named. Current naming contained within WP CDRL 13-011 (SI Document) is an extract from DISA. Unique identifiers will be assigned to each CI to facilitate item identification and version control. Version control of CIs is managed by CM at the Business Process or requirement level. The Requirements Traceability Matrix (RTM) identifies each requirement and specification to the associated CI.

System configuration settings will be documented in Phire and ADO as notes to the configuration task. The following information shall be recorded every time a change is made to configuration settings:

- Name of the person responsible for making the change (who)
- Date of change (when)
- Details of the change that was made (what)
- Reason for the change (i.e., a detailed description) so that the change can be referenced at a later date(why)
- Reference to any issues (e.g., issue number, cross-reference to a supporting position paper, etc.)

3.3 Configuration and Change Control

Configuration control is to maintain complete visibility over all changes to all configuration baselines through the entire system lifecycle. This approach ensures that the baseline(s) are always current and correct, that any proposed change has been reviewed for impact on baseline maintenance, and that all stakeholders have full visibility of all changes and of any potential impact to the system.

Configuration control will be exercised throughout the IPPS-A lifecycle to ensure software and hardware compatibility and to provide an orderly process to manage proposed changes. (See *IPPS-A Change Management Plan*)

3.3.1 Change Management System

A change management system includes the storage media, procedures, and tools that are used to record and access change requests.

Operation of the repository includes periodic auditing against current status of accounting information to ensure that the repositories contain all current baseline information. Internal audits will be scheduled, planned, and executed under the leadership of the IPPS-A Configuration Manager, but with the assistance of other areas of the organization.

Table 3-3 depicts the IPPS-A governance of the DRB, and TCB processes. The CM system includes the storage media, the procedures, and the tools that provide management and user access to configuration items and related information.

Table 3-3 IPPS-A Governance Structure

Type of Meeting	Type of Change	Approval
TCB	ECRs. Baseline establishment, Release recommendation, and changes/customization. Spec content and changes, System Configuration Interfaces. Recurring Infrastructure changes (network and system), patches, escalates to the EAB for conflict resolution	Baselines and approves changes to Allocated Baseline, and Internal Schedule
DRB	SPRs, SRs, ECR-Lows	Approves recurring normal routine changes with no cost/schedule or performance impacts

3.3.1.1 Types of Changes

Regardless of the type of change being requested, all change requests will be documented in ADO.

The following types of changes are applicable:

- Defects
  - Typically, system changes necessary to resolve a failure, malfunction, or data issues identified through Help Desk escalation and submitted through CRM
- Requirement Changes
  - Major Enhancements (ECRs) – a change that introduces new or updated business processes, when completed, will impact cost, schedule, or performance for an existing or new requirement. This also includes additions, modifications, and/or enhancements to existing IPPS-A capability, such as information exchanges and any law, regulation, or policy (LRP) changes. This also includes vendor service pack releases (i.e. Oracle, Gradkell, etc.) and DoD security mandates/patches.
  - Minor Enhancement (ECR-Lows) – a change that, when completed, will not impact cost, schedule, or performance of an existing requirement. This includes fixes that originate from the IPPS-A SI (e.g., code updates, configuration setting changes, training documentation updates, etc.)
- Code Changes – Production code changes migrated through the IPPS-A System for both PeopleSoft and Non-PeopleSoft applications
- Technical/Infrastructure Change – changes to the tools (ADO, Phire, Stonebranch, etc.), infrastructure, security, IAVAs, etc.
- Documentation Change – Documentation that is identified as a CI
- Production Release – Release package for defect fixes, enhancements, etc. in the production environment

Release types, as stated in the ChMP and RMP, determine the timing of the release implementation. All release types need to be evaluated for optimal timing and leveraging cutover activities. This enables the delivery to be achieved on the proper frequency to address the specific needs the efforts should fulfill. The release types are Normal, Urgent / Out of Cycle, and Emergency The release type categorizations are as follows:

- Normal (day-to-day business) o Severity 2 and below
  - Normal is assigned when an urgent / Out of Cycle or emergency release is not applicable, required, or justifiable
  - Scheduled in advance in accordance to release schedule
- Urgent / Out of Cycle (defect addressed within 5 business days)
  - Scheduled with as much advance notice as possible or in the next scheduled release

    - Data Fixes are not candidates for Urgent Release Requests
- Emergency (defect addressed within 24 hours)
    - Severity 1 and Severity 2; critical priority defects only
    - Authorized by an Emergency Change but still must be documented by the next business day

3.3.1.2 Engineering Change Requests

The ECR process is shown in Figure 4-1 The ECR form must be completed with all required data to allow a technical and cost analysis of the change for board approval.

Software migration ECRs are required for the production environments. These migrations are vetted and approved through the TCB process and Collab. Configuration changes to the baselined environments require ECRs. These changes are vetted and approved through the TCB Process and escalated to the Collab when Cost, Schedule, or Baseline changes are required.

The SI must notify the Government immediately in writing via email of an emergency or urgent requested change. The TCB Chair convenes a board meeting, virtually or physically using the preliminary notification documentation as the source document. If the board determines an emergency change is required, a preliminary ECR is requested, and its approval handled. All other priorities are handled through the CR Process as shown in Table 3-4.

3.3.1.3 Implementation of Configuration Control

The CM Team is responsible for implementing a change management process that ensures regulation of the flow of proposed changes, documentation of the complete impact of the proposed changes, and release of only approved configuration changes into CIs and their related configuration documentation.

Table 3-4 ECR Progression

Order	Status	Definition	Used By
1.	New	Ticket is created by the Change Requester	SI Developer, PMO, FMD
2.	Submitted	The Change Requester submits the ticket.	SI Developer, PMO, FMD

3.	Pending TCB Review	SI, PMO reviews the CR, collects the Level of Effort (LOE) from the SI development and updates the ticket	SI CM, PMO
4.	TCB Review	TCB reviews the CR and determines its impact to the Baseline, updates and documents the ticket and changes status to Pending Collab (or approved for Infrastructure updates)	PMO CM, SI CM, FMD
5.	Approved	Ticket is approved by the TCB. This status applies to Infrastructure changes that require funding.	PMO CM, TCB
6.	CV Approved	This status applies to approved items during the PI Confidence vote	PMO CM
7.	TCB	After the BMD issues a contract modification to SI and authorizes them to work. At this point, the ECRs are Approved on Contract.	PMO CM
8.	Deployed to Production	Change is deployed to Production. SI developer updates the ticket to document any issues encountered or success of the deployment	SI
9.	Closed	Change is closed after it is reviewed by the PMO CM	PMO CM

3.3.1.4 IPPS-A Version Control

The SI CM Team will track and provide control for changes to source code. The SI will be using both Subversion and Phire to ensure build and version control throughout the lifecycle and to include the design and supporting processes into the SI's CM Plan upon successful implementation of the tools.

Within the SI's CM Plan, the following is discussed:

- DQP Schema and Golden Gate Changes
- Golden Gate Changes tracked with Phire Workflow. The ESB Change requests are categorized into Baseline Standard for code/configuration changes and Baseline Golden Gate data/configuration changes

To ensure a quality product before being accepted into the Production Environment, the PMO CM Team will review the following Checklists and Standard Operating Procedures from the various SI Teams:

- SI ESB Development Leads: Checklists or SOPs to ensure that object codes within a change are documented in Phire for the CSR is be ready to go to PLT
- SI ESB Test Leads: Checklists or SOPs to ensure that testing of code is ready to be implemented in PLT successfully
- SI OGG Admins: Checklists/SOPs to ensure that they are following the proper steps prior to implementation into the Production Environment, this includes ensuring that the various environments are stable and are consistent. If an environment is not stable and/consistent, these risks will be addressed at the Risk Management Meetings for Government's decision

By following these guidelines, the PMO CM Team will collaborate closely with the SI Teams to introduce an improved way of verifying and validating data code into production.

3.3.1.4.1 Version Control System Physical Environment

To facilitate a Version Control System (VCS) model, appropriate system architecture must be identified.

3.3.1.5 Documentation and Data Management

All documentation for IPPS-A is retained in the CM Library. Documentation may be available in both physical and electronic formats. Refer to Section 5.0 for details on CDRL and artifact naming, versioning, and submittal.

- **Program Management Data:** Includes all formal program management data to include strategies, plans, processes, and procedures, as defined in DoDI 5000.88. Additional program management data includes charters, briefings, minutes, and official correspondence
- **Functional Proponent Data:** Official versions of documentation developed by FMD

- **CDRL Products:** All products developed under contract are Government property. Non-CDRL products developed under contract are also retained in the virtual and physical CM Library. CDRLs are delivered in accordance with the Contract Deliverable Process and Procedures.

Table 3-5 CDRL Products

Data Type	Product	Tool	Configuration Control
Requirements	Business Case, Functional Baseline, Allocated Baseline, RTM	ADO	Changes approved by applicable program control board. Includes waivers and deviations
Design	Enterprise Architecture design artifacts, velocity scripts	Cameo	Changes to baselined artifacts are approved by the applicable program board
Risk	Risk artifacts	IRIS	All Risks evaluated and mitigated through the Risk Management Control Board (RMCB)
Configuration Change	ECRs,	ADO	All changes approved by applicable program board
Test & Evaluation (T&E)	Test scripts, Test data, TIRs, SPRs	ADO	All artifacts approved by TRB
Operations and Support	Help Desk tickets	CRM ADO	Tracking for all tickets

Data Type	Product	Tool	Configuration Control
Software Configuration Management (SCM) for PeopleSoft	PeopleSoft source code	Phire	Manage source code throughout the lifecycle
SCM (non-PeopleSoft)	Objects, Exported Files, Scripts, Graphs, Binary files, Backup Files, Requirements, Digital Signatures	Ab Initio, OBIEE, GRC, DB sign (Gradkell)	Manage non-PeopleSoft source code
CI Tracking	CI data imports from tools	Service Now (Discovery)	Manage CIs
Data Acquisition	Data imports into encrypted staging databases	Oracle utilities, Ab Initio, Oracle ESB	Manage monthly data from source systems
CDRLs	Documentation	SharePoint	All Program documents and CDRLs

3.4 Configuration Status Accounting

Configuration Status Accounting (CSA) is the process of creating, organizing, retaining and reporting on the knowledge base necessary for the performance of CM. SI CM Team will maintain in CM databases that include as-designed, as-built, as-delivered, or as-modified configuration of all CIs, which is managed in the SI CM Plan.

The Service Now (Discovery) tool will collect metadata from the disparate program tools to automate most status and reporting functions. In addition to facilitating CM, CSA provides a highly reliable source of configuration information to support all program

and project activities including program management, systems engineering, software development, infrastructure, logistic support, and system modification and maintenance. The SI CM Team will generate and provide the following to the Release and Sustainment Managers for review:

- Monthly Software Configuration Status Accounting Reports (CDRL 13-010):

Provide detailed identification of object level changes reported through Service Now. Metrics derived from these reports are presented monthly at the SI Monthly Program Review (MPR). This report can also be run at any point during software development. Development and maintenance of this report is the responsibility of the SI and approved by the Government

- **Quarterly CSAI Reports (CDRL 13-010)**: Provide a complete inventory of IPPS-A, including software, hardware, documentation as configured, and a listing of outstanding ECRs, ECPs, and SPRs. Development and maintenance of this report is the responsibility of the SI and approved by the Government

Status Accounting: Provides status information on requirements, SPRs, ECRs, ECPs, and test status. This information is retained in the requirements, change and test modules of ADO, Stat, Subversion, and other tools described in APPENDIX B

3.4.1 ECR Reporting

Each ECR is retained in ADO and tracked and reported through each step-in acceptance, development, test, and implementation. This information is available at any time through the tool. Actions and decisions from each DRB, TCB, and Collab are logged at the individual ECR or ECP level in the tool and are immediately available.

3.5 Configuration Verification and Audit

The purpose of a configuration audit is to verify and document that a CI and its defined Configuration Identification, as specified in baseline documentation, are accurate, complete, and have met total project needs. An audit is an evaluation to ascertain compliance to specifications, standards, agreements, or other criteria. In CM, there are two types of audits:

- Functional Configuration Audits (FCAs)
- Physical Configuration Audits (PCAs)

Periodically, SI CM Team will perform physical and functional configuration audits. They will be conducted to confirm the implemented IPPS-A configuration is consistent with configuration requirements and baseline documentation. The timing of periodic audits will typically be coordinated with releases of new versions of the system.

The SI CM Team, in conjunction with the Audit Team, will conduct or oversee configuration audits to ensure that the baseline and CIs comply with established specifications, drawings, and other relevant material, thereby guaranteeing confidence in the configuration

integrity of the system. Such audits will ensure that the CIs reflect the functional and physical characteristics defined in the corresponding documentation, thus validating the accuracy and comprehensiveness of the documents.

The CM Team will use the audit procedure to review the internal change control process, ensuring that each step of the process is functioning as originally designed. The following section provides an overview of the responsibilities and practices for conducting audits of the IPPS-A system and system components.

3.5.1 Typical Audit Actions

The actions associated with audits are usually divided into three parts: pre-audit, audit, and post-audit. Each of these is covered separately in the material that follows.

In these audit actions, the role assigned to the CM Team can vary and may include management of the audit, co-chairing one or more of the audits themselves, and acting as recording secretary or clerk. The view taken here is that the CM Team is responsible for all the actions associated with the audit, including both pre-audit and post-audit actions. The reasons for this are:

1) Complete documented authority and responsibility for such audits are not established
2) Personnel who are otherwise engaged in areas for which they have responsibility will not usually divert efforts from their primary tasks to perform other jobs
3) Successful completion of these audits is of direct interest to the PMO and SI CM Teams

3.5.2 Configuration Audit Roles and Responsibilities

This section provides an overview of the roles and responsibilities that the various organization areas have in the planning, preparation, and conduct of the audits.

3.5.2.1 PPS-A Management Roles and Responsibilities

The IPPS-A PdM has the authority and responsibility for the following:

- Authorization of schedule and execution of configuration audits
- Allocation of sufficient resources and time for the audit activities
- Approval of Audit Reports

This authority and responsibility may be delegated.

3.5.2.2 CM Team Roles and Responsibilities

The SI CM Team will supervise the activities of the Configuration Audit Team. The SI CM Team will also have the following roles and responsibilities:

- Ensure that the audits are planned, scheduled, and that the results are reported to the TCB
- Coordinate participation on the Audit Team
- Upon completion of the audit, track the identified deficiencies through to correction and resolution, and then submit the Production Baseline to the TCB for approval and formal establishment
- Publish Audit Reports and recommendations
- Establish a repository for historical configuration information to be used as the approved standard for future audits

3.5.3 Verification

The IPPS-A PMO T&E Branch supports the CM function by verifying that all baselined requirements have been met by the design and development of IPPS-A.

The Redline Process streamlines correction and closure of documentation errors discovered during test and eliminates multiple formal submittals. The Redline Documentation Process ensures that documentation errors identified during test are corrected, verified, and closed expeditiously to facilitate the test schedule and applies to all documents to be tested. CM manages the process throughout the test phases.

Verification activities are performed against current approved baselined requirements through test and evaluation. The T&E Branch verifies the Allocated Baseline. CM provides identification and control of the baselines, environments, and test artifacts. Specific CM activities that support T&E are discussed in the *IPPS-A Test Management Plan.*

3.5.4 Internal Audits

3.5.4.1 CM Library Content

Audits of the content of the CM Library (including documentation in the Requirements,

Configuration Change, and Test repositories) will be performed semi-annually, by both the CM Librarian and the QA function to ensure that all expected programmatic and SI deliverable items are present and under version control. For more information, refer to Section 5.0.

3.5.4.2 Change Control Process

CM will audit the change control processes and databases to ensure that the changes are accurately documented and appropriate actions recorded.

3.5.4.3 Baseline Configuration Audits

CM is responsible for oversight of the SI's Baseline Configuration Audits (BCAs) performed by the SI CM staff against a Government approved SI Audit Plan. The audits are accomplished in coordination with Government Test for the software through automated compare processes The BCA Plan is approved by the CM Manager to ensure that the current, comprehensive artifacts representing the configured system have been identified and accepted. Subsequent BCAs may be held immediately before and after each test event to ensure there have been no undocumented changes made to the system.

The BCA is performed by the SI CM staff against the approved Audit Plan. The plan identifies specifically what is to be included in the audit. The Audit Report includes the hardware and infrastructure topology, network diagrams, and identification, by name and serial number of all hardware associated with IPPS-A, and all software running on the identified IPPS-A hardware.

The report identifies what exists against the approved baseline, and any additions or subtractions that have been made since the previous audit.

3.5.5 System Verification Review

The SI CM Team is responsible to perform a System Verification Review (SVR), the formal examination of functional characteristics of CIs, prior to Government acceptance, to verify the system or identified CIs have achieved the requirements specified in the functional and allocated configuration documentation. An SVR must be conducted prior to the operational test and FDD for each IPPS-A release at the Government's discretion.

SVRs are performed by the SI using an Audit Plan approved by the Government to ensure that all necessary artifacts are included. The audit is facilitated by the SI's CM organization. The audit team includes Government and SI members. Government acceptance of the Configuration Audit Report, including the audit checklist, concludes the specific audit being held. The Audit Checklist is an appendix to the specific Audit Plan and is used to establish the Audit Plan and Report.

3.5.5.1 Functional Configuration Audit and Physical Configuration Audit

The SI CM Team is responsible for hosting, facilitating, and reporting on the SVR and

FCA/PCAs. Reports for these audits are done against audit plans for each type of audit. MIL-HDBK 61-A provides an audit plan template as a guide. The plans are submitted to and approved by the Government as defined in the CDRL items.

3.5.6 Configuration Change Control – Object and Code Level

The SI CM Team is responsible for identifying, versioning, and controlling the individual Enterprise Resource Planning (ERP) and third-party application internal and external objects within the available object management tool.

3.5.7 Configuration Change Control – Infrastructure Detail Level

The Government is responsible for change control to the infrastructure and environments through the applicable program control boards. Artifacts representing the complete topology of the system are approved after the initial BCA certifies that IPPS-A physical configuration matches the design artifacts.

Configuration Audits will typically be conducted by the SI CM Team either annually or immediately following each major release of the IPPS-A production system. At a minimum, an audit will occur at least once every 18 months following a major release.

4.0 CM LIBRARY

This section applies to the configuration control of all IPPS-A documentation and artifacts. The procedures and guidance for the management of the CM Library and its intended use by IPPS-A PMO and the SI detailed in this section.

4.1 CM Library Roles and Responsibilities

The IPPS-A PMO is ultimately responsible for the management and maintenance of the CM Library. Overall, the Configuration Management Library will include:

- Maintain all applicable configuration baseline documentation
- Assist with the development and maintenance of the processes and procedures for CM library, in accordance with, the Army Records Information Management System (ARIMS) and assist with access controls availability.
- Maintain and record copies of all program documentation and contract deliverables.
- Maintain records of transmission and receipt of all Government Furnished Information (GFI).
- Maintain records of all product baselines and Configuration Items (CIs).

4.2 CM Document Identification and Handling

Each document has a unique identifier in SharePoint so that it can be correctly associated with the applicable configuration (product identifier and revision) of the associated item, referred to precisely, and retrieved when necessary.

The CM Library processes discussed in this plan must be followed to submit a PMO or CDRL document to the library.

Documents should be requested from the CM Librarian for any updates or changes.

4.3 CM Library Content, Storage, Processes

The following subsections describe the library contents, electronic and physical storage, and the applicable processes for CDRL and artifact submittals. The CM Library uses AR 25-400-2 as a guide for record retention and destruction.

4.3.1 Library Content

The following types of documents are stored in the CM Library to include, but not limited to the following types of records:

- **Program and Milestone Documents:** Include Acquisition Decision Memorandums, test plans, and program baselines, etc., and are version controlled and stored in the CM Library
- **Government Furnished Information (GFI) Documents:** Officially requested to be delivered to an outside organization and typically delivered in a package with a record of delivery
- **Reference Materials:** Created or acquired documents that are preserved solely for reference or exhibition purposes, to include extra copies of documents preserved only for convenience and reference in the physical library
- **Regulations, Policies, and Laws:** Provide guidance and direction for the IPPS-A program to follow and are utilized during document creation to ensure compliance
- **CDRL Documents:** CDRL documents are due to the Government from any group (Support Contract, System Integrator, etc.) that has a contract with the Government. CDRLs are listed in a contract with specific dates of delivery to the Government. CDRLs are maintained in the CM Library's Deliverables site and use the file naming and versioning described in Section 5.2 as a guide but may have additional descriptors.
- **Meeting Minutes/Recordings:** Details of IPPS-A PMO meetings to include recordings if applicable

4.3.2 Electronic and Artifact Documentation Submission Process.

To submit a document to the CM Library, type the following URL in your web browser:

CM Library - Default

The CM Librarian will verify the information and ensure the document is properly routed.

The owner of each document/artifact is required to follow the submission process to ensure the document is baselined and stored. The owner must identify if the document is draft or final, the version, and any required access restriction to the CM Librarian.

The submission process output is a baseline version of the submitted document. Each item is considered the formal document until a new version is submitted or other action is performed as specified by the document approver.

APPENDIX A LIST OF ACRONYMS AND ABBREVIATIONS

Acronym	Definition
ACM	Application Change Management (Phire)
ADO	Azure DevOps
AEA	Army Electronic Archives
ANSI	American National Standards Institute
ARIMS	Army Records Information Management System
BCA	Baseline Configuration Audit
BMD	Business Management Division
CDD	Capability Development Document
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CFI	Cross-Functional Integration
CI	Configuration Item
CM	Configuration Management
CMP	Configuration Management Plan
COR	Contracting Officer's Representative
COTS	Commercial off-the-Shelf
CSA	Configuration Status Accounting
CSAI	Configuration Status Accounting Information
CSR	Change/Service Request
DD	Design Document (Formerly DFCDD, TDD)
DFCDD	Detailed Functional Configuration Design Document
DoD	Department of Defense
DoDI	DoD Instruction
DPdM DRB	Deputy Product Manager Defect Review Board
DVS	Data Validation Script
EA&I	Enterprise Applications & Integration
ECR	Engineering Change Request
ERP	Enterprise Resource Planning
ETL	Extract, Transform, Load
FD	Full Deployment
FDD	Full Deployment Decision
FMD	Functional Management Division
FRD	Functional Requirements Document
GAT	Government Acceptance Test
GFI	Government Furnished Information
GOSC	General Officer's Steering Committee

HCM	Human Capital Management
HR	Human Resources
IA	Information Assurance
IAVA	Information Assurance Vulnerability Alert
IBM	International Business Machines
IDD	Interface Design Document
IOC	Initial Operational Capability
IPPS-A	Integrated Personnel and Pay System – Army
IRS	Interface Requirements Specification
KO	Contracting Officer
LUT	Limited User Test
MIL-HDBK	Military Handbook
MS	Microsoft
NIPRNet	Non-secure Internet Protocol Router Network
OS	Operating System
OT	Operational Test
OT&E	Operational Test and Evaluation
OV	Operational Views
PCA	Physical Configuration Audit
Pdf	Portable Document Format
PdM	Product Manager
PEO E	Program Executive Office, Enterprise
QA	Quality Assurance
RFP	Request for Proposal
RHA	Records Holding Area
PM	Project Manager
RMCB	Risk Management Control Board
PMO	Project Management Office
RTM	Requirements Traceability Matrix
RTVM	Requirements Traceability Verification Matrix
SAAR	System Authorization Access Request (Form)
SCM	Source Code Management
SDP	Software Development Plan
SE&A	Systems Engineering & Architecture
SEIT	Systems Engineering, Integration & Test
SI	System Integrator
SIPRNet	Secret Internet Protocol Router Network
SLA	Service Level Agreement
SPR	Software Problem Report
SR	System Requirements
S/SS	System/Subsystem Specification

SV	System Views
SVD	System Version Description
SVR	Software Verification Review
SyRS	System Requirements Specification
TBAI	Technical and Business Architecture Integration
TCB	Technology Control Board
T&E	Test and Evaluation
TIRT&E	Test Incident Report Test and Evaluation
VMVCS	Virtual Machine Version Control System
VM	Virtual Machine

APPENDIX B CM TOOL DESCRIPTIONS

- **Azure DevOps (ADO):** PPS-A leverages AzureDevOps (ADO) to achieve alignment and efficiency in work processes. ADO houses the program's Backlogs, Roadmaps, Metrics, and various dashboards to monitor the full lifecycle of work from funnel to released. The tool is managed primarily by IPPS-A PMO with FMD and SI regularly providing updates and data to provide the program with high levels of accuracy on progress.
- **Microsoft (MS) SharePoint:** Used for document delivery collaboration, document repository, CDRL workflow, and CM Library. MS SharePoint supports the workflow for IPPS-A PMO CM from user submission to requirement addition or change and allocation to a release by release management. The ADO and SharePoint tools are used to obtain activity information for the PMO in providing the most accurate, timely, and easily accessible information. This information includes current status of any change, the history of any change, and the schedules for and status of verifications and audits, as well as resultant action items. The SI manages and the PMO TMD Team administers SharePoint.
- **Phire:** Used for migration of PeopleSoft code and configuration between databases and environments, as well as for tracking migration of third-party application custom code. Phire uses the Change/Service Request (CSR) to track the migration workflow. Manual migrations are tracked in Phire. A detailed status report from Phire identifying all PeopleSoft configuration changes and the current state of the objects under development can be generated at any time. The SI manages Phire for code migrations.

APPENDIX C WP CDRLS CM REVIEWS

- 13-009- Configuration Management Plan
- 13-010- Quarterly Configuration Status Accounting Information (CSAI) Report
- 13-011- Configuration Item Document Updates
- 13-015- Quality Assurance and Metrics Plan
- 13-016- Program Metric Reports
- 13-017- Release Notes